Using the two-dimensional loss landscape analysis, we explore the landscape of cosine similarity $F(\alpha, \beta)$ between an original image and its perturbed version in the embedding space of DINOV2, where two independent noises drawn from the standard Gaussian distribution with their respective scaling factor α and β are added to the pixel values of the original image. Figure 4 shows the average landscape of cosine similarity over 100 real and AI-generated images, respectively, where the real ones are sampled from ImageNet, and the AI-generated ones are generated by the ablated diffusion mode (ADM) (Dhariwal & Nichol, 2021). It can be observed that real images attain a higher similarity after perturbation than AI-generated images. The reason is that DINOV2 is trained only on real images and their data augmentations, which makes its embedding space less sensitive to noise perturbations on real images. Following (He et al., 2024), we compare AEROBLADE and RIGID on three datasets consisting of real and AI-generated images: ImageNet, LSUN-Bedroom, and DF40 (deepfakes) (Yan et al., 2024). The AI-generated images are collected from a variety of generative models, including different variants of diffusion models, generative adversarial networks (GANs), and variational autoencoders (VAEs). Table 2 reports their AUROC scores, where higher values indicate better detection performance. RIGID is shown to outperform AEROBLADE. The difference can be explained by the fact that AEROBLADE relies on the pre-trained autoencoder of an LDM for detection, which may have limited generalization ability for images generated by different models. On the other hand, RIGID is agnostic to image generation models.

AI-generated Text Detection via Adversarial Learning Due to the nature of autoregressive and probabilistic sampling of next-token prediction in LLMs, statistical approaches that derive statistics from LLMs for AI-generated text detection are popular baselines. Examples include log-likelihood (log p), token ranks (rank), and predictive entropy (entropy). Mitchell et al. propose DetectGPT (Mitchell et al., 2023), a detection method that adopts a log-likelihood ratio test between the original text and its perturbed-and-reworded versions. However, recent studies have shown that many AI-generated text detectors are vulnerable to AI paraphrasing (Sadasivan et al., 2025) – using an LLM to rewrite the original AI-generated text can circumvent the reliability of many detectors. To address this challenge, Hu et al. (Hu et al., 2023) propose a detection method called RADAR, which uses adversarial learning to iteratively train a detector and a paraphraser to improve the robustness of AI-generated text detection. During RADAR's training, the paraphraser uses proximal policy optimization to update its parameters, using the detector's predictions about its paraphrased text as a reward. The detector takes human-written text, original AI-generated text, and paraphrased AI-generated text as input, to maximize detection performance on human and AI-generated text. This iterative training process continues until detection performance is saturated on a validation dataset. With this adversarial learning setup, the detector learns to be robust in detecting both original and paraphrased AI-generated text.

Following the same experimental setup as in (Hu et al., 2023), we collect human-written and AI-generated text from four different text datasets, including text summarization, question answering, Reddit's writing prompts, and TOFEL exam essays, where 8 different LLMs are used for text generation. In the evaluation, we also use OpenAI's GPT-3.5-Turbo API as a paraphraser to test the robustness of detectors. This paraphraser was not used to train any of the detectors. In addition to the above detectors, we include OpenAI's RoBERTa-based detector[3] for comparison. Table 3 compares the detection performance of different methods using the AUROC score. In the scenario of no paraphrasing, many detectors can perform well, achieving AUROC scores as high as 0.904. However, when tested against paraphrasing, all detectors show a significant degradation in detection performance (up to more than 50% decrease), except for RADAR. The stability of RADAR's performance can be attributed to its use of adversarial learning for robust detection.

6. Discussion: When AGI Means Artificial Good Intelligence

This section discusses several insights to accelerate research and innovation in safety from a signal processing perspective. While the trends of GenAI lean toward the mindset of creating frontier AI technology by moving fast and scaling everything, especially for frequent model/product releases and intensified training and inference resources, embracing boldness and imperfection is becoming the new norm for AI safety. However, we argue that the development and use of scientific methods and tools, such as signal processing, is the most important milestone that must be achieved to claim an undisputed victory in GenAI. In particular, the computational AI safety framework introduced in this paper can be further developed into a new discipline that encompasses a comprehensive list of topics such as safety risk exploration (e.g., benchmarks, red-teaming, adversarial testing, and attacks), safety risk mitigation (e.g., detection, model updates, and safety-enhanced training), and evaluation (e.g., risk-capability analysis, safety certification, human-centered interaction, and governance). By studying and practicing computational AI safety, we believe we are just at the beginning of a new era and a bright future towards the ultimate goal of building a *Safety Generalist* to "use AI to govern AI," which involves autonomous exploration, mitigation,

[3] https://huggingface.co/openai-community/roberta-base-openai-detector

Table 3. AUROC scores of AI-generated text detectors under two scenarios: *without paraphrasing* and *with paraphrasing* on the original AI-generated text. The paraphraser is OpenAI's GPT-3.5-Turbo API. Except for RADAR, all detectors show a performance drop when detecting paraphrased AI-generated text.

Scenario	log p	rank	log rank	entropy	DetectGPT	OpenAI	RADAR
Without Paraphrasing	0.887	0.755	0.904	0.472	0.864	0.900	0.856
With Paraphrasing	0.442	0.461	0.426	0.651	0.434	0.627	0.857

and evaluation of safety risks with human oversight as a controllable knob. The methods and techniques discussed in this paper only scratch the surface of signal processing for computational AI safety, and open science and community support are the organic engines to accelerate the momentum.

We don't (and shouldn't) need to build safety guardrails from scratch for GenAI While GenAI may seem like a new technology, many lessons and past experiences learned from safeguarding pre-GenAI machine learning (ML) systems can be used as a foundation to quickly build safety guardrials for GenAI. This kind of "knowledge transfer" can avoid the mistakes of reinventing the wheel and effectively shorten the development of safety tests and patches for emerging GenAI systems. In addition, as the context and model size of GenAI continue to grow, building safety guardrails from scratch is not only time-consuming, but also resource-intensive. As an example, Table 4 shows a one-to-one mapping between the research topics in adversarial robustness (Chen & Hsieh, 2023) for pre-GenAI machine learning systems and the seemingly new safety challenges in GenAI. The problems therein can be formulated using the same principles, and therefore the methods and tools developed for classical ML can be easily extended to study safety in GenAI. Specifically, adversarial examples stem from finding problematic data inputs with minimal and human-imperceptible changes to cause misclassification of the target ML system, while jailbreak features prompt modification and stealth to induce harmful output for GenAI. Data poisoning and backdoor are training-phase risks that are directly related to biased or malicious instruction tuning and model fine-tuning for GenAI. Out-of-distribution generalization in ML explores the robustness under semantically similar but unseen changes during training, while alignment in GenAI means teaching the model to learn the values and principles implicit in the alignment data and generalize beyond the provided context. Finally, model reprogramming (Chen, 2024) explores cross-domain ML by attaching input and output transformation functions to a weight-frozen ML model, similar to prompt injection in GenAI, which manipulates a target GenAI system to produce an incompliant output via prompt hacking.

Open Challenges and Opportunities To strengthen the link between signal processing and AI safety, we highlight some open research challenges and opportunities.

- Computational safety for multi-modal GenAI: GenAI technology is on its way to accepting inputs from multiple data types and potentially generating outputs with mixed modalities. While multi-modal GenAI may introduce increased safety risks by providing additional attack surfaces, techniques such as heterogeneous (multi-view) signal processing applied to multiple data sources can be explored for safety research.
- Computational safety for agentic and physical GenAI: The next generation of GenAI is to be empowered with the ability to act on behalf of the user by autonomously creating agentic workflows and potentially participating in multi-agent interactions. In addition, GenAI is expected to thrive in embodied agentic systems (e.g., AI humanoid robots), extending the discussion of AI safety from the digital space to the physical world (Tang et al., 2024). Control-based signal processing methods may be a promising direction to explore agentic and physical safety for GenAI.
- Practical AI safety guardrails: In addition to addressing the inherent trade-offs between safety and capability, the practical implementation of AI safety guardrails must be cost-effective in terms of computation, memory, and agility. Compression and high-dimensional statistics in signal processing can be powerful tools to address long-context and real-time requirements.
- Understanding the fundamental limits of AI safety: To better understand and design alignment training and reward modeling in GenAI, theoretical signal processing approaches can be used to characterize the information-theoretic limits, statistical properties, or computational optimality for GenAI. Furthermore, model-based signal processing can be explored as a potential approach to overcome the limitations of GenAI.

May AGI Mean "Artificial Good Intelligence" Although AGI is often used as shorthand for Artificial General Intelligence, and there are many expectations that AGI may be a near-term possibility, we argue that computational safety is a critical prerequisite for the development and deployment of safe and responsible AI technology. Through

Table 4. The mapping between the topics of adversarial robustness in classical (pre-GenAI) machine learning systems and the seemingly new safety challenges in GenAI. Many techniques developed for classical machine learning can be easily extended to study the associated safety challenges in GenAI.

Classical Machine Learning	GenAI
Adversarial Example	Jailbreak Prompt
Data Poisoning / Backdoor	Biased or Malicious Instructions
Out-of-Distribution Generalization	Alignment
Model Reprogramming	Prompt Injection

the presented computational AI safety framework, this paper's primary objective is to elucidate the pivotal role of signal processing in the examination and resolution of numerous foundational and pragmatic safety concerns pertaining to GenAI. We posit that this paper can serve as a catalyst for researchers and practitioners to accord greater consideration to signal processing techniques as a means of "good intelligence." Should AGI remain the overarching objective, it is of the utmost importance to ensure that this term represents "Artificial Good Intelligence." Moreover, while safety is becoming the new arms race among leading GenAI stakeholders, we strongly believe that *safety should not be a competition and a race to the bottom*. Instead of a winner-take-all game, AI safety should be considered a public good through open science and borderless collaboration. After all, the true worth of AI safety is as useful and meaningful as the individuals who can benefit from it.

7. Conclusion

This paper presents a computational safety framework for GenAI through the lens of signal processing. We show that many problems that arise in AI safety can be formulated in a unified hypothesis testing scheme. In particular, we study two representative use cases in AI safety, jailbreaks and AI-generated content, and elucidate how signal processing techniques can be used to build reliable detectors and develop efficient mitigation strategies. Our results suggest the effectiveness of hypothesis testing as a fundamental methodology for improving AI safety. Finally, we articulate our views on the sustainability and growth of AI safety research and technology adoption, and why signal processing, a foundational technique for pattern recognition and machine intelligence, is a promising approach to accelerating its progress.

Acknowledgment

The authors thank Mr. Xiaomeng Hu and Mr. Zhiyuan He for their help in providing the materials for the figures and numerical results.

References

Anil, C., Durmus, E., Rimsky, N., Sharma, M., Benton, J., Kundu, S., Batson, J., Tong, M., Mu, J., Ford, D. J., et al. Many-shot jailbreaking. In *Neural Information Processing Systems*, 2024.

Bereska, L. and Gavves, E. Mechanistic interpretability for AI safety–a review. *arXiv preprint arXiv:2404.14082*, 2024.

Chao, P., Robey, A., Dobriban, E., Hassani, H., Pappas, G. J., and Wong, E. Jailbreaking black box large language models in twenty queries. *IEEE Conference on Secure and Trustworthy Machine Learning*, 2025.

Chen, P.-Y. Model reprogramming: Resource-efficient cross-domain machine learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pp. 22584–22591, 2024.

Chen, P.-Y. and Hsieh, C.-J. *Adversarial Robustness for Machine Learning*. Elsevier, 2023.

Chua, J., Li, Y., Yang, S., Wang, C., and Yao, L. AI safety in generative AI large language models: A survey. *arXiv preprint arXiv:2407.18369*, 2024.

Dhariwal, P. and Nichol, A. Diffusion models beat gans on image synthesis. *Advances in neural information processing systems*, 34:8780–8794, 2021.

He, Z., Chen, P.-Y., and Ho, T.-Y. RIGID: A training-free and model-agnostic framework for robust ai-generated image detection. *arXiv preprint arXiv:2405.20112*, 2024.

Hsu, C.-Y., Tsai, Y.-L., Lin, C.-H., Chen, P.-Y., Yu, C.-M., and Huang, C.-Y. Safe LoRA: the silver lining of reducing safety risks when fine-tuning large language models. *Advances in Neural Information Processing Systems*, 2024.

Hu, X., Chen, P.-Y., and Ho, T.-Y. RADAR: Robust AI-text detection via adversarial learning. *Advances in Neural Information Processing Systems*, 36:15077–15095, 2023.